Reconstruction of Missing Indoor Temperature Sensor Measurements Using Numerical Interpolation Methods

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1. Engineering Problem

Indoor temperature monitoring is important for building environmental assessment, thermal comfort evaluation and HVAC system analysis. However, sensor faults, communication failures and data transmission issues may cause missing temperature measurements, resulting in incomplete environmental records.

This project investigates the reconstruction of missing indoor temperature measurements using numerical interpolation and regression-based methods. The objective is to evaluate how accurately different computational approaches can estimate missing temperatures and how their performance changes with missing-data duration. The reconstructed values are intended to support retrospective analysis of incomplete sensor records rather than future temperature forecasting.

The study focuses on one selected indoor sensor and missing intervals of approximately 10–120 minutes. The results may not generalise to other buildings, extended outages or conditions where surrounding measurements are unavailable.

2. Inputs and Outputs

2.1 Dataset Description

The project uses an indoor environmental sensor dataset containing timestamped temperature, humidity and atmospheric pressure measurements. Sensor ID `6012002000869` was selected, producing 13,800 complete, unique observations after preprocessing, covering February to June 2024.

Temperature was the reconstruction target, while humidity and pressure were used as additional regression predictors.

Variable	Mean	Standard deviation	Observed range
Temperature (°C)	23.09	1.16	19.0 – 28.5
Relative humidity (%RH)	45.03	7.39	26-75
Pressure (mb)	1017.83	5.66	1002 - 1028

Table 1. Summary statistics for the 13,800 cleaned sensor measurements.

The nominal sampling interval was approximately 10 minutes, although irregular intervals were observed. Actual timestamps were therefore retained for numerical reconstruction.

2.2 Data Preprocessing

Raw sensor records were processed by extracting environmental measurements, converting Unix timestamps into datetime values, and converting measurements into numerical formats. Duplicate timestamps were checked, incomplete records were removed, and the remaining observations were chronologically sorted.

The dataset was divided chronologically into 80% training data (11,040 observations) and 20% testing data (2,760 observations). The regression models were fitted using the training subset, while reconstruction performance was assessed on the held-out testing subset.

To evaluate reconstruction accuracy, selected known temperatures were artificially hidden while their original values were retained as ground truth.

2.3 Computational Inputs and Outputs

The computational inputs include timestamps, temperature measurements and, for regression-based approaches, humidity and pressure. Numerical interpolation uses known temperatures surrounding a missing interval, while endpoint-constrained regression additionally requires reliable temperature values at both boundaries.

The primary output is estimated temperature (°C) at missing timestamps. Additional outputs include validation-window records, error metrics and comparison figures.

Incomplete environmental records were excluded during preprocessing. The window-generation function also checks selected input parameters and skips candidate windows without sufficient surrounding measurements. Reconstruction remains unsupported when essential predictors or boundary measurements are unavailable.

3. Computational Solution

Seven reconstruction approaches were investigated. The initial evaluation compared six methods; an endpoint-constrained regression approach was subsequently developed after examining the initial results.

Method	Main principle	Required information
Persistence	**Retain last known temperature**	**Previous temperature**
Linear interpolation	**Straight-line reconstruction**	**Time and two endpoints**
Cubic spline	**Smooth piecewise cubic interpolation**	**Surrounding temperatures**

Humidity regression	**Predict temperature from humidity**	**Humidity**
Pressure regression	**Predict temperature from pressure**	**Pressure**
Combined regression	**Predict temperature from two variables**	**Humidity and pressure**
Endpoint-constrained regression	**Correct regression using known endpoints**	**Regression predictors and two endpoints**

Table 2. Summary of the seven reconstruction approaches.

3.1 Numerical Interpolation and Baseline

Persistence was selected as a baseline because it provides the simplest reconstruction by maintaining the last known temperature throughout the missing interval.

Linear interpolation estimates temperature using two surrounding known measurements:

$$T(t) = T_L + [(t - t_L)/(t_R - t_L)](T_R - T_L)$$

where T_L and T_R represent the observed temperatures at the left and right boundaries, respectively.

Cubic spline interpolation was implemented using SciPy's **CubicSpline** with natural boundary conditions. Four known measurements before and four after each missing interval were used to construct a smooth piecewise cubic curve.

Linear interpolation is computationally simple but cannot represent complex changes within a gap. Cubic splines provide greater flexibility but may introduce overshooting or unrealistic intermediate values.

3.2 Regression Models

Three Ordinary Least Squares regression models were implemented using scikit-learn's LinearRegression: humidity-only, pressure-only and combined regression.

The combined model is:

$$T_reg(t) = \beta_0 + \beta_H\, H(t) + \beta_P\, P(t)$$

where H and P represent humidity and pressure, and β₀, β_H and β_P are fitted regression coefficients.

Regression was investigated to determine whether environmental predictors could provide additional reconstruction information. However, these models assume that the relationships

between temperature and the predictor variables remain sufficiently consistent across the training and testing periods.

Unlike interpolation, the original regression predictions do not explicitly satisfy known temperature values surrounding a missing window.

4. Validation and Verification

Artificial missing-temperature windows were created within the held-out testing data. The hidden temperature values were retained as ground truth for evaluating reconstruction accuracy.

Missing duration	Number of windows
10 minutes	**15**
30 minutes	**15**
60 minutes	**15**
120 minutes	**15**
Total	**60**

Table 3. Artificial missing-temperature validation windows.

Each reconstruction method was evaluated using the same 60 windows. The final seven-method comparison therefore contained 420 method-window evaluations.

Three error metrics were used:

- **MAE (°C):** Mean absolute difference between actual and reconstructed temperatures.
- **RMSE (°C):** Square root of the mean squared error, giving greater weight to large deviations.
- **NRMSE (%):**Normalises RMSE by the testing temperature range, allowing errors to be evaluated relative to temperature variation. This helps avoid interpreting a small RMSE as necessarily indicating good performance when temperature variation is limited (Khoshvaght et al., 2025).

$$NRMSE\ (\%)\ =\ RMSE\ /\ (T_max, test\ -\ T_min, test)\ \times\ 100$$

The testing temperature range was 19.5–25.5°C, giving a normalisation denominator of 6.0°C. Reported mean NRMSE values were calculated by averaging the errors across the validation windows for each method and duration. The performance of each reconstruction method is compared in Sections 5 and 7.

Implementation correctness was also checked using automated unit tests covering missing-gap generation, interpolation, error calculations, reconstruction evaluation and endpoint constraints. All eight existing tests passed.

The validation approach allows reconstruction errors to be compared against known measurements. However, the artificial windows do not reproduce every possible condition of a real sensor failure.

5. Initial Six-Method Results

The initial evaluation compared persistence, linear interpolation, cubic spline interpolation and three regression models.

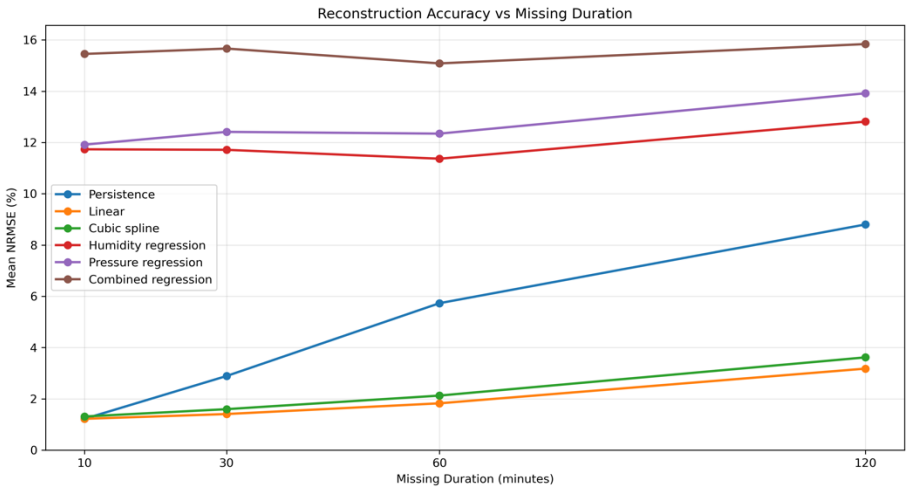


Figure 1. Mean NRMSE of the initial six reconstruction methods across four missing-data durations.

The results showed that numerical interpolation generally produced substantially lower errors than the regression models. Linear interpolation performed particularly well for short and medium missing intervals, while persistence errors increased as the missing duration became longer.

Regression was initially expected to provide useful additional information through humidity and pressure. However, all three regression models produced relatively large reconstruction errors. The combined regression performed especially poorly, with mean NRMSE values around 15–16% across the investigated durations.

Further examination of reconstructed temperature curves revealed that the regression predictions often failed to match the known temperatures at the boundaries of missing windows. This highlighted a limitation of using environmental predictors without considering the surrounding temperature measurements.

These observations motivated the development of an improved regression reconstruction method incorporating known boundary temperatures.

6. Method Improvement: Endpoint-Constrained Regression

The original combined regression model estimated temperature from humidity and pressure without enforcing agreement with the known measurements immediately before and after a missing window.

To improve boundary continuity, an endpoint-constrained correction was introduced.

First, a time-dependent weighting factor was defined:

$$\alpha(t) = (t - t_L)/(t_R - t_L)$$

The corrected reconstruction was then calculated as:

$$T_EC(t) = T_reg(t) + (1 - \alpha)[T_L - T_reg(t_L)] + \alpha[T_R - T_reg(t_R)]$$

The first term represents the original regression prediction. The remaining terms provide corrections based on the prediction errors at the known left and right boundaries.

This adjustment ensures that the reconstructed curve passes through both observed endpoint temperatures.

Importantly, the method does **not** reduce the number of fitted regression coefficients or change the underlying regression model. Instead, it constrains the final reconstruction by applying a boundary-based correction.

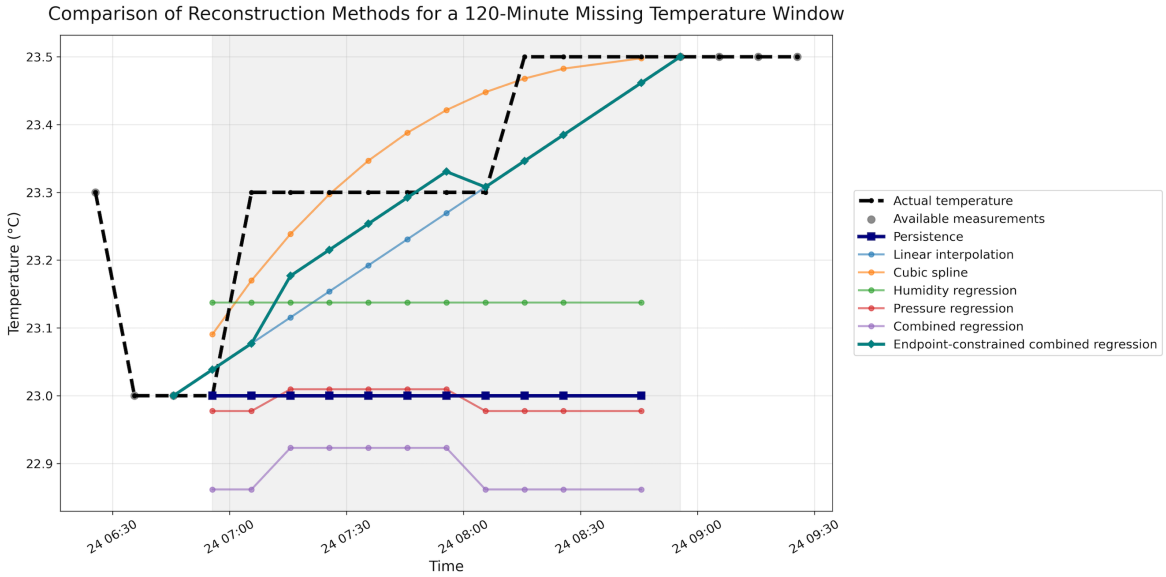


Figure 2. Comparison of seven reconstruction methods for a representative 120-minute missing temperature window.

Figure 2 illustrates that the original regression predictions do not closely follow the surrounding temperature measurements, whereas the endpoint-constrained reconstruction connects the known boundaries and captures the general temperature trend.

Nevertheless, matching the boundaries does not guarantee accurate predictions within the missing interval. The corrected method was therefore evaluated quantitatively against the initial six approaches.

The new method also requires temperatures at both boundaries and available humidity and pressure data. Consequently, it is suitable for retrospective reconstruction but cannot directly replace real-time forecasting.

7. Final Results and Discussion

7.1 Seven-Method Comparison

The endpoint-constrained method was evaluated using the same validation windows as the initial six methods.

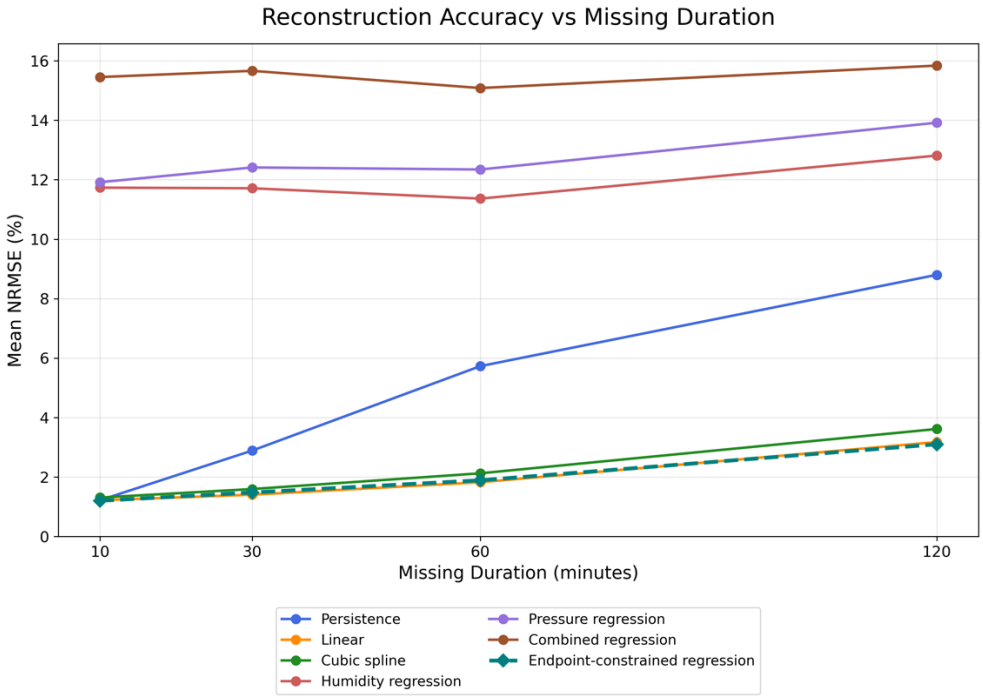


Figure 3. Mean NRMSE comparison of seven reconstruction methods across missing-data durations.

Reconstruction method	10 min	30min	60 min	120 min
Persistence	**1.222**	**2.8907**	**5.7316**	**8.8016**
Linear interpolation	**1.2216**	**1.4075**	**1.8261**	**3.1792**

Cubic spline	**1.3119**	**1.5979**	**2.1272**	**3.6176**
Humidity regression	**11.7334**	**11.7125**	**11.3655**	**12.8124**
Pressure regression	**11.9157**	**12.4124**	**12.3442**	**13.9177**
Combined regression	**15.4526**	**15.6619**	**15.0829**	**15.8383**
Endpoint-constrained regression	**1.2039**	**1.4825**	**1.8920**	**3.1035**

Table 4. Mean window-level NRMSE (%) for all seven methods. Lower values indicate better reconstruction accuracy.

7.2 Effectiveness of Endpoint Constraints

The endpoint-constrained approach substantially reduced reconstruction error compared with the original combined regression.

For 120-minute gaps, mean NRMSE decreased from 15.8383% to 3.1035%, representing an approximately **80.4% relative reduction**.

The constrained method achieved the lowest mean NRMSE for the 10-minute and 120-minute gaps. However, linear interpolation remained slightly more accurate for the 30-minute and 60-minute gaps.

These results indicate that applying known temperature boundary information greatly improves the original regression-based reconstruction. Nevertheless, the improvement cannot be attributed entirely to humidity and pressure, because the corrected method also uses observed temperature endpoints.

7.3 Comparison and Engineering Interpretation

Linear interpolation remained highly competitive despite its simplicity. This suggests that, for the tested sensor and missing intervals, the temporal information from surrounding temperature measurements was more useful than the relationships provided by humidity and pressure alone.

Cubic spline interpolation produced relatively low errors but generally did not outperform linear interpolation. Its additional flexibility therefore did not provide a consistent accuracy advantage for this dataset.

Persistence was reasonable for very short gaps but deteriorated with longer missing durations, demonstrating the limitations of assuming constant temperature.

The endpoint-constrained regression improves continuity by incorporating known boundary temperatures, but requires additional computations compared with linear interpolation.

To evaluate computational efficiency, four reconstruction methods were benchmarked across the same 60 missing windows, with 20 repeated runs. The median runtime was recorded for each method. Linear interpolation achieved the shortest runtime (0.0659 ms), followed by cubic spline interpolation (5.2052 ms), combined regression (17.9064 ms), and endpoint-constrained regression (63.7809 ms), as presented in Table 5.

Although endpoint-constrained regression achieved slightly lower mean NRMSE for 120-minute gaps (3.1035% compared with 3.1792% for linear interpolation), this improvement was relatively small compared with its additional computational cost. This suggests that linear interpolation provides a more favourable accuracy–efficiency trade-off for the investigated dataset. However, the measured runtimes were small in absolute terms and excluded data preprocessing and model training. The results are also dependent on the hardware and implementation used.

The constrained method was developed after examining the initial results and was assessed using the same validation windows. Consequently, its performance should be interpreted as an improvement within the investigated dataset rather than evidence of general superiority on completely independent data.

Method	Median runtime for 60 windows (ms)
Linear interpolation	**0.0659**
Cubic spline	**5.2052**
Combined regression	**17.9064**
Endpoint-constrained regression	**63.7809**

Table 5. Median reconstruction runtime across 60 validation windows, based on 20 repeated runs per method. Data preparation and model training were excluded.

8. Limitations and Conclusion

This study was limited to one indoor sensor, four missing-data durations and artificially created missing windows. The results may differ for other sensors, buildings, operating conditions or longer missing periods.

Interpolation-based methods depend on suitable surrounding temperature measurements, while regression requires available environmental predictors. A complete sensor outage may remove both types of information and prevent reliable reconstruction.

Despite these limitations, the project demonstrated that relatively simple numerical techniques can reconstruct short periods of missing indoor temperature data effectively.

The initial six-method comparison revealed poor performance from humidity- and pressure-based regression. Introducing endpoint constraints substantially reduced the original combined regression error, producing accuracy comparable to linear interpolation.

However, the results also showed that a more complex approach does not necessarily provide a meaningful improvement over a simple numerical method. For the investigated dataset, linear

interpolation offered a particularly effective balance between accuracy, simplicity and data requirements. Future work could investigate additional sensors, longer missing intervals, independent validation datasets, and end-to-end runtime and memory performance.

9. AI Use and Critical Reflection

The research direction and engineering problem were independently selected by the author. After defining the project objectives and identifying the methods to investigate, ChatGPT was used extensively to generate Python code and support implementation.

AI contributed substantially to code development, including data preprocessing, interpolation, regression models, validation, debugging, testing and documentation. However, the generated code was executed and reviewed against actual sensor measurements and experimental results.

After observing poor regression performance, I investigated its limitations and used AI to assist with implementing endpoint constraints. AI-generated explanations were also critically evaluated, including correcting the misconception that endpoint constraints reduced the regression model's degrees of freedom.

Overall, AI improved programming efficiency and supported my understanding of numerical methods, while research decisions, result interpretation and verification remained my responsibility.

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